PROJECT REPORT

TASK 17: DATASET EXPLORATION USING PANDAS

1. Abstract

This project explores the Iris dataset using Python and Pandas. It examines the dataset's dimensions, column names, data types, numerical summary statistics, missing values, duplicate rows, and species distribution.

The project aims to develop an understanding of a dataset's structure and basic characteristics before performing further analysis or building a machine-learning model.

2. Introduction

Dataset exploration is an important step in data analysis and machine learning. Before building a model, it is necessary to understand what information the dataset contains, how many observations are available, which variables are numerical or categorical, and whether data-quality issues exist.

Pandas provides several functions for these initial checks, including `head()`, `tail()`, `shape`, `columns`, `info()`, and `describe()`.

This project applies these functions to the Iris dataset using Google Colab.

3. Objectives

The objectives are:

- Load a dataset into a Pandas DataFrame.
- Display its first five and last five records.
- Identify the number of rows and columns.
- Inspect column names and data types.
- Generate numerical summary statistics.
- Identify numerical and categorical variables.
- Check for missing values and duplicate rows.
- Record at least five observations about the dataset.

4. Tools and Technologies

Python: Programming language used for analysis.

Pandas: Used to load, inspect, and summarize tabular data.

NumPy: Used for numerical data operations.

Scikit-learn: Used to access the built-in Iris dataset.

Google Colab: Browser-based environment used to execute and document the notebook.

GitHub: Used to store the notebook and supporting project files.

5. Dataset Description

The project uses the Iris dataset, a classic dataset used for introductory data analysis and classification exercises.

The dataset contains 150 observations and five columns in the DataFrame created for this project.

The four numerical features are:

- Sepal length
- Sepal width
- Petal length
- Petal width

The fifth column, species, identifies the Iris species associated with each observation.

There are three species: setosa, versicolor, and virginica.

The dataset reference is:
https://archive.ics.uci.edu/dataset/53/iris

6. Methodology

The project follows these steps:

1. Load the Iris dataset using Scikit-learn.
2. Convert it into a Pandas DataFrame.
3. Add the species names as a categorical column.
4. Display the first five records using `head()`.
5. Display the last five records using `tail()`.
6. Inspect dimensions and column names.
7. Examine data types and non-null counts using `info()`.
8. Generate descriptive statistics using `describe()`.
9. Identify numerical and categorical columns.
10. Check missing values and duplicate rows.
11. Examine the number of observations in each species group.
12. Record at least five observations from the results.

7. Pandas Functions Used

7.1 `head()`

Displays the first five rows by default and helps inspect the beginning of the

dataset.

7.2 `tail()`

Displays the last five rows by default and helps inspect the end of the dataset.

7.3 `shape`

Returns the number of rows and columns in the DataFrame.

7.4 `columns`

Provides the list of column names.

7.5 `info()`

Shows the column names, non-null counts, data types, and memory usage.

7.6 `describe()`

Provides numerical summary statistics such as count, mean, standard deviation, minimum, quartiles, and maximum.

7.7 Missing-value and duplicate checks

`isnull().sum()` counts missing cells in each column, while `duplicated().sum()` counts duplicate rows.

8. Dataset Exploration Results

The exploration identifies the following characteristics:

- The DataFrame contains 150 observations and five columns.
- There are four numerical measurement columns and one categorical species column.
- The dataset contains three species, with 50 observations per species.
- The supplied dataset has no missing cells.
- Descriptive statistics summarize the distribution and spread of numerical measurements.
- The mean petal length differs across species, making this feature useful to investigate when studying species separation.

Insert the actual output tables and findings from your executed Colab notebook in this section.

9. Key Observations

Observation 1: Dataset dimensions

The dataset contains 150 rows and five columns, providing 150 observations for exploration.

Observation 2: Numerical and categorical variables

Four columns contain numerical measurements, while the species column represents a categorical label.

Observation 3: Species distribution

The dataset contains setosa, versicolor, and virginica, with 50 observations per species.

Observation 4: Missing values

The missing-value check reports no missing cells in the provided dataset.

Observation 5: Numerical variation

The descriptive statistics show that the features have different ranges and standard deviations. Comparing these values helps reveal which measurements vary more within the dataset.

Observation 6: Differences in petal length

The species-wise petal length summary shows distinct average values across the three species, especially between setosa and the other species.

10. Importance of Dataset Exploration

Dataset exploration supports informed decisions before modelling. It helps analysts understand the data structure, identify missing values, examine feature distributions, and determine which variables may be useful for further analysis.

It also helps identify potential data preparation steps before training a machine-learning model.

11. Limitations

This project focuses on basic dataset inspection and descriptive analysis. It does not train a predictive model, perform advanced feature engineering, or evaluate model performance.

12. Future Enhancements

The project could be extended with visualizations such as histograms, scatter plots, and box plots. Further analysis could explore correlations among numerical features and compare distributions across species.

A later stage could use the explored dataset to build and evaluate a classification model.

13. Conclusion

The project demonstrates the use of Pandas for initial dataset exploration. By examining dimensions, column names, data types, descriptive statistics, missing values, duplicate rows, and species distribution, the project builds a foundation for more advanced data analysis and machine learning.

14. References

1. Pandas documentation: https://pandas.pydata.org/docs/
2. Scikit-learn dataset documentation: https://scikit-learn.org/stable/datasets/toy_dataset.html
3. UCI Iris dataset: https://archive.ics.uci.edu/dataset/53/iris

15. Project Links

GitHub Repository: https://github.com/venky1318/Dataset_Exploration